THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 11: **Electricity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of resistivity is: 1

(a) ohm per metre **(b)** ohm metre **(c)** ohm **(d)** volt

2. The number of electrons in 1 coulomb of charge is about: 1

(a) 6.25×10^{18} **(b)** 6.25×10^{19} **(c)** 1.6×10^{-19} **(d)** 1.6×10^{19}

3. Two resistors of 6 Ω each in parallel give a resistance of: 1

(a) 6 Ω **(b)** 3 Ω **(c)** 36 Ω **(d)** 12 Ω

4. A student states: “The resistance of a wire depends on the material.” The correct response is: 1

(a) Correct **(b)** Only on length

(c) Incorrect **(d)** Cannot be determined

5. For the power of a device using 2 A at 220 V, Student A says 440 W; Student B says 110 W. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The instrument used to measure potential difference is: 1

(a) galvanometer **(b)** ammeter **(c)** rheostat **(d)** voltmeter

7. The slope of a V–I graph gives: 1

(a) power **(b)** current **(c)** charge **(d)** resistance

8. Work done in moving 2 C of charge through 12 V is: 1

(a) 14 J **(b)** 24 J **(c)** 6 J **(d)** 0.17 J

9. The commercial unit of electrical energy is: 1

(a) joule **(b)** watt **(c)** volt-ampere **(d)** kilowatt hour

10. An electric bulb rated 60 W, 220 V draws a current of about: 1

(a) 0.27 A **(b)** 3.67 A **(c)** 13 200 A **(d)** 0.6 A

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is meant by electric current? Write its SI unit. 2

12. Why is tungsten used for filaments of bulbs? 2

13. Calculate the current through a 40 Ω resistor connected to a 12 V battery. 2

14. A student says that the resistance of a conductor increases with an increase in the area of cross-section. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A 6 Ω and a 3 Ω resistor are connected in parallel, and this combination is in series with a 4 Ω resistor across a 12 V battery. Find the total current. 3

16. An electric bulb is rated 100 W, 220 V. Find 3
(i) its resistance
(ii) the current drawn
(iii) energy consumed in 10 h.

17. Why does the cord of an electric heater not glow while its element does? 3

OR

Find the resistance of a copper wire 1 m long with area 1×10^{-6} m² ($\rho = 1.6 \times 10^{-8}$ Ω m). 3

18. What is electric power? Write three formulae for power and its SI unit. 3

OR

How many 176 Ω resistors in parallel are required to carry 5 A on a 220 V line? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe an experiment to verify Ohm's law with a circuit diagram. 5
(ii) Draw the expected V–I graph.
(iii) What does its slope represent?

OR

(i) What are the disadvantages of series connection for domestic circuits? 5
(ii) Three bulbs of 4 Ω, 8 Ω and 8 Ω are connected in parallel across 8 V. Find the total current and current in each.

20. **Case study:** A student connected two identical bulbs first in series and then in parallel to the same battery. 5
(i) In which case is total resistance more?
(ii) In which case do the bulbs glow brighter?
(iii) If each bulb has 10 Ω and the battery is 10 V, find the current in each case.
(iv) What happens if one bulb fuses in each case?

OR

(i) Derive the expression for heat produced: $H = I^2Rt$. 5
(ii) An electric kettle of resistance 50 Ω draws 4 A. Find the heat produced in 1 minute.

END OF QUESTION PAPER — **Total: 40 Marks**

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